

PARKTOWN BOYS' HIGH SCHOOL



Department of Mathematics

Project

27 January 2026

Subject	Mathematics
Grade	12
Topic	Functions and Inverses
Marks	50
Time	1 hour

Examiner

Ms H Gouws

Moderator

Mr J Odendaal

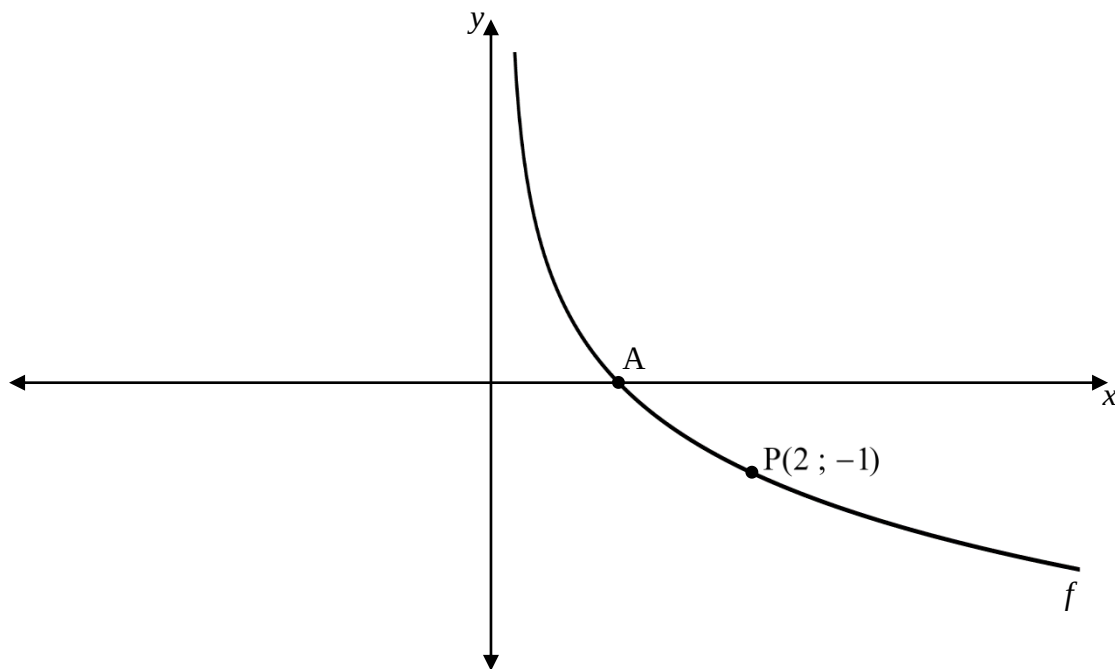
INSTRUCTIONS

Read the following instructions carefully before answering the questions.

1. This question paper consists of 4 tasks and 7 pages, including an answer sheet.
- Detach the answer sheet, write your name on it and hand it in with your answers.
2. Clearly show ALL calculations, diagrams, graphs, etc. that you have used in determining your answers.
3. Diagrams are NOT necessarily drawn to scale.
4. Number your questions exactly like the questions in the question paper.
5. Answers only will NOT necessarily be awarded full marks.
6. You may use an approved scientific calculator (non-programmable and non-graphical), unless stated otherwise.
7. If necessary, round off answers to TWO decimal places, unless stated otherwise.
8. Use the mark allocation to guide the length of your solution.
9. Write neatly and legibly.
10. Rule off after each question.

TASK 1

The graph of $f(x) = \log_{\frac{1}{2}}(x)$ has been sketched below. $P(2 ; -1)$ is a point on the graph.

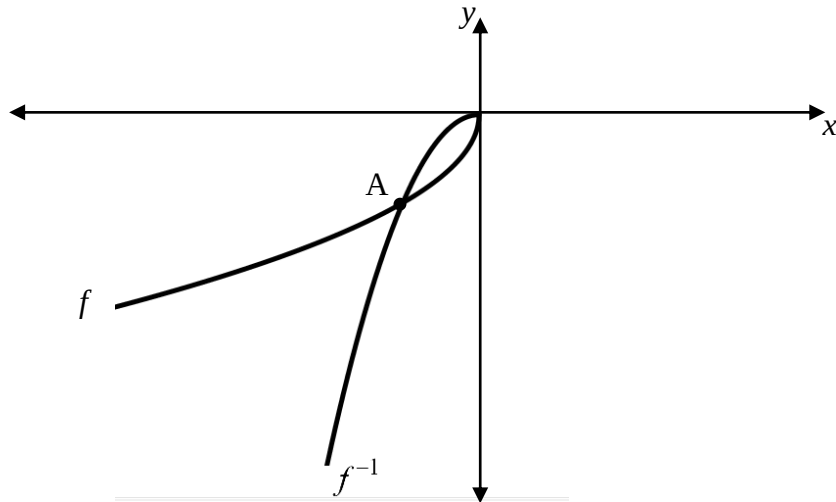


- 1.1 Write down the domain of f . (1)
- 1.2 Write down the coordinates of A . (2)
- 1.3 Write down the range of f^{-1} . (1)
- 1.4 Sketch each of the following graphs on the set of axes provided. Clearly label your graphs and indicate the coordinates of the image of A and the image of P on your sketches.
 - 1.4.1 $g(x)$, the reflection of f in the y -axis. (5)
 - 1.4.2 $h(x)$, the reflection of f in the line $y = x$. (5)
- 1.5 For which value(s) of x is $f^{-1}(x) > 2$? (2)

[16]

TASK 2

In the diagram below, the graph of $f(x) = -\sqrt{-\frac{1}{3}x}$ and f^{-1} has been sketched.

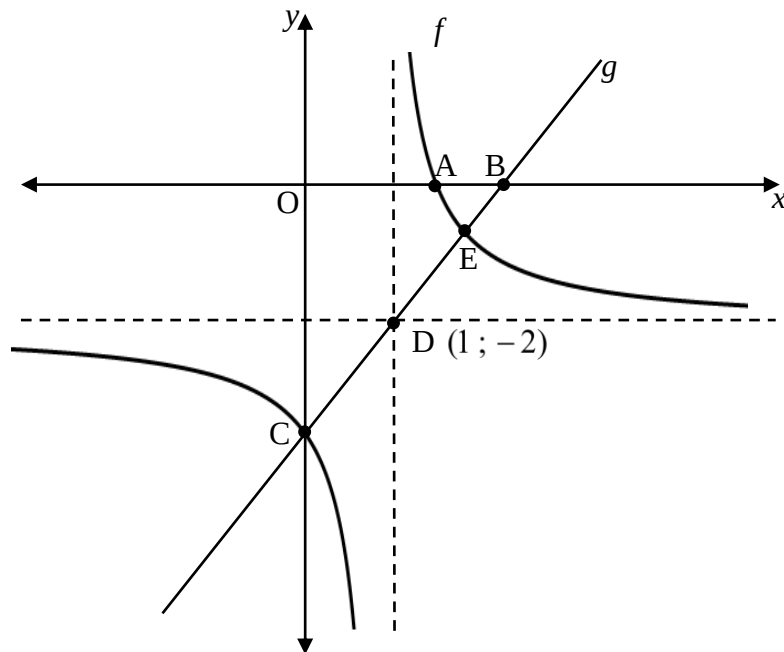


- 2.1 Write down the equation of the line of symmetry of f and f^{-1} . (1)
- 2.2 Determine equation f^{-1} . (3)
- 2.3 Determine the coordinates of A. (3)
- 2.4 g is obtained by reflecting f^{-1} around the line $y = 0$. Determine the equation of g . (3)
- 2.5 f^{-1} is shifted 1 unit right and q units up to form h . If h intersects $j(x) = -6x + 3c$ at the point $(1 ; 2c)$, determine the value of q . (5)

[15]

TASK 3

In the diagram below $f(x) = \frac{a}{x+p} + q$ and g , a line of symmetry of f , have been sketched.



- 3.1 Show that the coordinates of C is $(0; -3)$. (2)
- 3.2 Show that $f(x) = \frac{1}{x-1} - 2$. (3)
- 3.3 Calculate the length of AB. (4)
- 3.4 Determine the coordinates of E. (5)
- 3.5 For which value(s) of x is
- 3.5.1 $f(x) \geq 0$? (2)
- 3.5.2 $f(x) > g(x)$? (3)

[19]

NAME: _____

CLASS: _____

TASK 1.4

